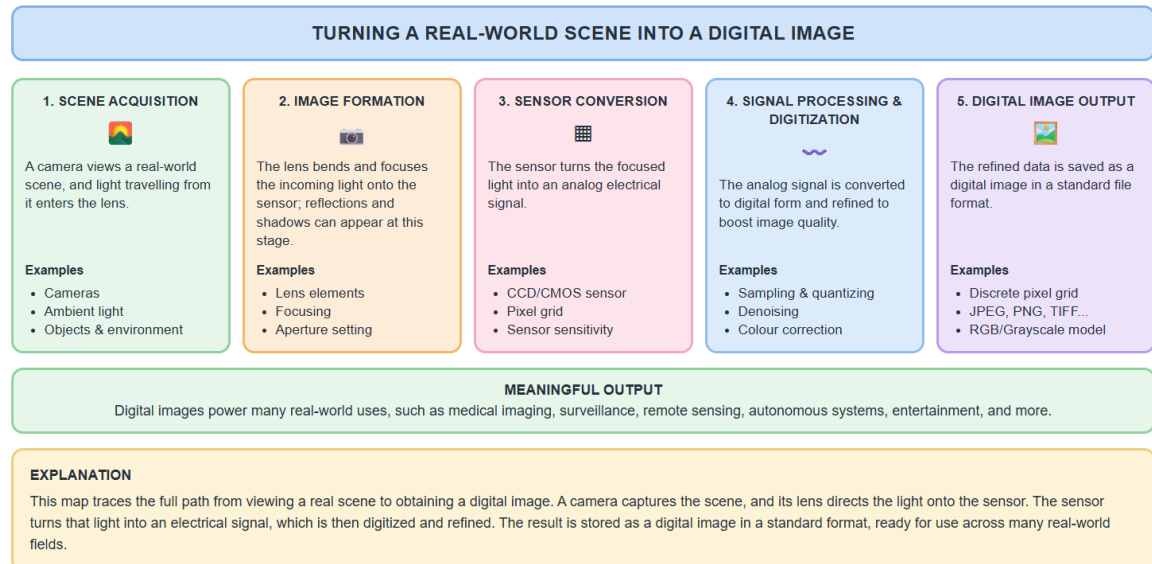
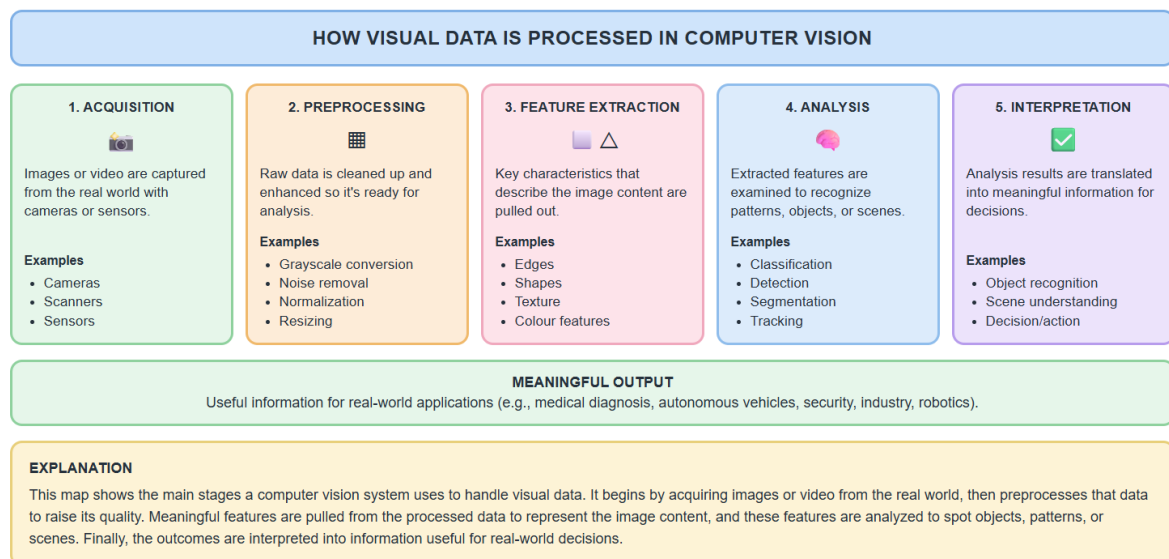


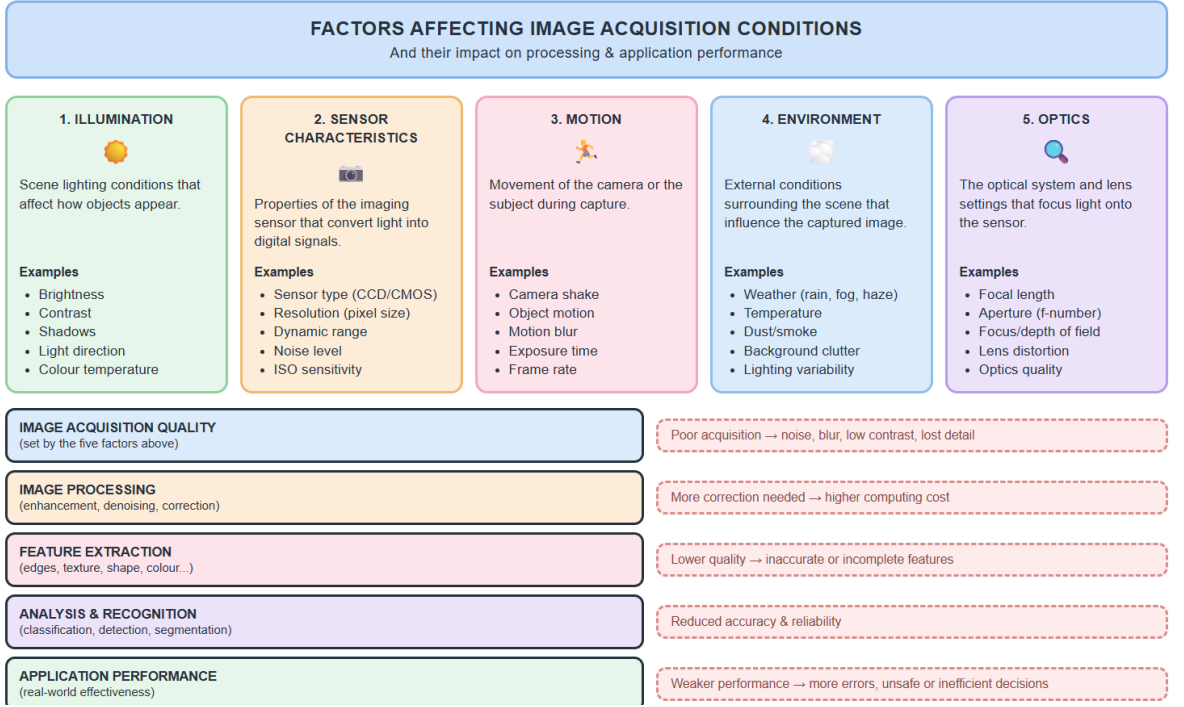
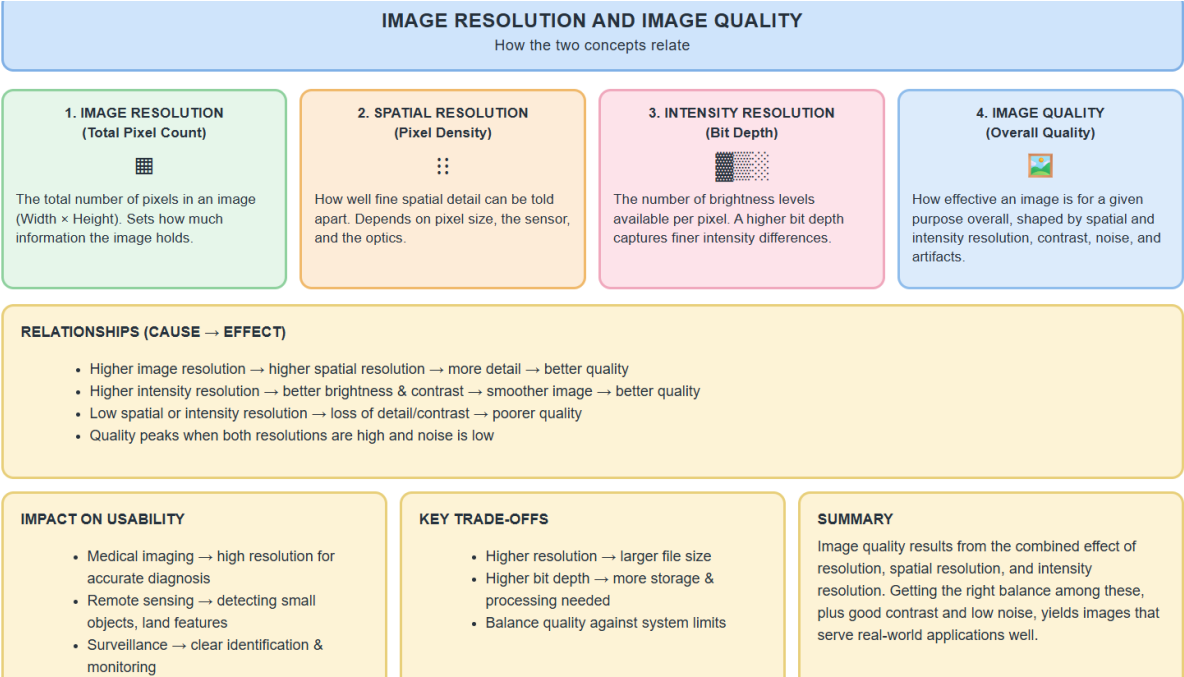
# ASSESSMENT TOOL 2– Concept Mapping

1.



2.





## COMPUTER VISION SYSTEM CAPABILITIES AND CHOOSING AN APPLICATION

### MAJOR SYSTEM CAPABILITIES

#### 1. IMAGE ACQUISITION



Captures images or video at good quality and resolution.

**Effect on applications**  
High-quality input enables accurate results.

#### 2. PROCESSING



Enhances images and manages noise, lighting, and distortion.

**Effect on applications**  
Improves reliability under real-world conditions.

#### 3. FEATURE EXTRACTION



Detects and pulls out key elements such as edges, shapes, textures.

**Effect on applications**  
Shapes how well objects, patterns, or regions are identified.

#### 4. ANALYSIS & RECOGNITION



Classifies, recognizes, detects, and interprets visual content.

**Effect on applications**  
Directly drives the accuracy and efficiency of outcomes.

#### 5. REAL-TIME PERFORMANCE



Processes data quickly and responds without delay.

**Effect on applications**  
Essential for time-critical uses like surveillance and robotics.

#### 6. RELIABILITY & ROBUSTNESS



Performs consistently across conditions and handles uncertainty.

**Effect on applications**  
Ensures steady performance in safety-critical applications.

### GUIDES SELECTION OF THE RIGHT VISION-BASED APPLICATION

#### EXAMPLE APPLICATIONS

- Medical imaging
- Industrial inspection
- Autonomous vehicles
- Surveillance & security
- Agriculture monitoring
- Robotics & automation

#### EXPLANATION

A computer vision system relies on several capabilities working together to deliver meaningful results. How strong each capability is shapes how well the system performs across different applications. Weighing these capabilities helps in picking the most suitable vision-based application and reaching better accuracy, efficiency, and reliability.